

Question	Answer	Marks
3(a)(i)	$E = \sigma / \varepsilon$ or $E = \text{gradient}$	C1
	$E = \text{e.g. } 12 \times 10^7 / 0.0050$ $= 2.4 \times 10^{10} \text{ Pa}$	A1
3(a)(ii)	cross drawn at (1.0%, $24 \times 10^7 \text{ Pa}$), labelled Q	B1
3(b)	resultant force (in any direction) is zero	B1
	resultant moment / torque (about any point) is zero	B1
3(c)(i)	(moment =) $33 \times 0.65 / 2$ or $1.5 \times (0.65 - 0.12)$ or $T \sin 50^\circ \times (0.65 / 2)$	C1
	sum of clockwise moments = sum of anticlockwise moments $33 \times (0.65 / 2) + 1.5 \times (0.65 - 0.12) = T \sin 50^\circ \times (0.65 / 2)$	C1
	tension = 46 N	A1
3(c)(ii)	$\sigma = F / A$	C1
	$\pi r^2 = 46 / (1.5 \times 10^7)$ $r = 9.9 \times 10^{-4} \text{ m}$	A1

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3(c)(iii)	elastic limit is not reached or (new) stress is less than (stress at) elastic limit or (new) strain is less than (strain at) elastic limit	M1
	(so the wire behaves) elastically	A1

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